



Editorial

Launch of the UK SABR Consortium Pelvic Stereotactic Ablative Radiotherapy Re-irradiation Guidelines and National Audit

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As clinical oncology trainees in the 2000s we were taught that re-irradiation was to be avoided at all costs due to concerns about serious toxicity; we were only exposed to extracranial stereotactic ablative radiotherapy (SABR or SBRT) by travelling to visit international colleagues. Fast forward a couple of decades and pelvic SABR re-irradiation has recently been commissioned by NHS England for routine use. This has been made possible through many years of diligent work by a large number of colleagues.

Pelvic SABR re-irradiation was initially used within NHS England as part of the SABR Commissioning through Evaluation (CtE) programme. CtE ran for 3 years and allowed us to treat patients while collecting key data points. The aim was to test a range of pre-defined clinical end points, ahead of a full commissioning decision. At the outset, the aim was to achieve 1-year overall survival and local control of 60% and 50%, respectively, with grade 3 and grade 4/5 toxicity of <20% and <5%, respectively [1]. With limited follow-up, the 1-year overall survival was not reached. However, the calculated probability was reported as 92%. The 1-year local control rate was 76% and grade 3 toxicity was 3.8%, with no documented episodes of grade 4/5 toxicity. The authors also showed cost-effectiveness by comparing it to pelvic exenteration. Following the CtE report, pelvic SABR re-irradiation was commissioned in England in January 2021 [2].

So, how do we approach the routine delivery of pelvic SABR re-irradiation following CtE? There are no high-level evidence-based data to guide us, particularly with respect to planning and organ-at-risk (OAR) dose volume constraint

calculations. A systematic review by Murray *et al.* [3] reviewing the small retrospective series published reported they were enormously heterogeneous in nature, limiting any useful conclusions. This lack of evidence was also highlighted in the recently published Delphi consensus, where practising clinicians were polled on different aspects of the technique [4]. Although there was a 92% consensus on issues of set-up and delivery, only 50% consensus was reached on questions regarding OAR dose volume constraints. Finally, the lack of data was reflected in the heterogeneity of OAR dose volume constraint calculations used within CtE. A survey of CtE clinicians revealed different techniques were used in different centres and for different patients in the same centre. These included use of baseline OAR dose volume SABR constraints as for a radiation-naïve pelvis; allowing for 30–50% recovery from the initial radiation dose to OARs [3,5]; a cumulative constraint as per historic spinal cord data by Nieder *et al.* [6,7]; and a maximum cumulative constraint published by Abusaris *et al.* [8]; finally, others calculated the dose they wished to be delivered to the target and worked back to identify the recovery required to allow that. As a result, identifying the optimal approach is challenging.

There is an effort underway by the University of Leeds to utilise the CtE data further. DARIUS (Dose Accumulation in Re-Irradiation Using Science) is a retrospective study funded by CRUK RadNet seed funding aiming to gain access to treatment plans from several centres. They aim to model potential dose volume constraints in the pelvic re-irradiation setting. In an ideal world, we would design a prospective trial to study this relatively novel treatment modality, but even if we were to apply for funding tomorrow, a trial would be years away from reporting. Additionally, designing a trial in this setting is challenging, as power calculations based on single-centre retrospective

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cohorts with a few dozen patients is not favourably viewed by funders; and there are many confounding factors, including multiple systemic anticancer therapies.

In the UK we have an opportunity to gather data in a way that sits between retrospective single-centre data and fully-funded prospective clinical trials. NHS clinicians are exceptionally good at collaborating to collate large datasets, as shown in the CtE programme and national audits. Previous Royal College of Radiologists (RCR) audits, run by the RCR Clinical Oncology Quality Improvement and Audit Committee (COQIAC), have been hugely successful and influenced local, national and international practice [9–12]. The RCR Anal Cancer Audit used data from conformal radiotherapy trials as the proven standard; a national anal intensity-modulated radiotherapy guideline was produced and implemented across the UK, before an RCR Anal Cancer Audit was carried out to quantify implementation of the guidance, cancer outcomes and toxicity. Following the audit, the UK Anal Guidance is now standard of care throughout the UK and being adopted as RCR Guidance. It informed the design and is the standard arm for the ongoing phase III PLATO study (Personalising Anal cancer RadioTherapy dose, ISRCTN88455282). The prophylactic nodal dose used in our UK national guidance is now being used internationally, for example in the ongoing German phase II RADIANCE trial (Radiochemotherapy with or without Durvalumab In the treatment of ANal squamous CELL carcinoma NCT04230759) [13].

We aim to take a similar approach with pelvic SABR re-irradiation. The UK SABR Consortium are producing guidance, as they do for all SABR indications. Following the implementation of this guidance and the wider implementation of pelvic SABR re-irradiation around the UK, the UK SABR Consortium, in collaboration with RCR COQIAC and the NHS clinicians delivering this service, aim to carry out a national prospective audit. We propose using the CtE data as the standard. This will run over 2 years and aim to quantify patterns of use around the UK, with outcomes including local control and survival, and toxicity. These data will be interrogated and reported by the treating clinicians with a minimum of one collaborator from each centre participating in the project. Development of the guidance and a subsequent audit form part of an iterative quality improvement project, to optimise pelvic SABR re-irradiation.

We have made three major changes from pelvic SABR re-irradiation delivered in CtE to optimise the treatment and data collection:

- (i) The introduction of more detailed guidance.

From our survey of clinicians delivering treatment during CtE, we are aware of wide variation of OAR constraint calculations, as detailed above. As Timmerman [14] so eloquently put it in his recent editorial regarding the initial dose volume constraints for lung SABR: ‘except for the spinal cord, all other limits for this original table were my educated guesses’. He detailed that a degree of pragmatism was required to introduce the technique, which then allowed

him to learn from where he started and adapt. While we acknowledge that the proposed pelvic SABR re-irradiation OAR dose volume constraint method is based on low-level evidence and recent experience of those treating within CtE, our aim was to produce pragmatic guidelines to ensure a consistent approach across participating centres to allow iterative learning and adaptation of the guidance. Development of these guidelines involved an initial small multidisciplinary group to scope and prepare the first draft, followed by circulation among stakeholders for comments, subsequent revisions, then finally a confidential advanced copy sent to a wider group for final comments [15].

- (ii) Centres will collect patient-reported outcome measures (PROMs), using EORTC QLQ-30 and the relevant tumour-specific questionnaire, at baseline and at 1 year.

There are no high-quality PROM data in pelvic SABR re-irradiation. In order to capture both general quality of life and organ-specific toxicities in a format that can tie together all the different tumour types included, following discussion with academics in the quality of life field, we have opted to use EORTC PROMs. We are limited by the timeframes of this audit to 1 year and have minimised collection points to two to maximise return at these time-points. These data will empower clinicians and patients to make informed decisions between pelvic SABR re-irradiation and alternative treatments in the future, by quantifying different aspects of quality of life following SABR.

- (iii) There will be an option for individualising dose prescription.

In the pelvic SABR re-irradiation setting, patients are treated with a variety of tumour types, indications, anatomy and previous radiation characteristics (dose, fractionation, location, intervening time period, etc). Additionally, we might have a different acceptable threshold of normal tissue complication, based on the clinical situation. For example, it may be reasonable to accept a different level of toxicity for an elderly patient with three positive nodes from prostate cancer with the aim to delay systemic therapy, than for a young patient with a solitary lateral sidewall node from rectal cancer with the aim to ablate as an alternative to surgery. In a series of 69 patients with locally recurrent rectal cancer (LRRC) treated with 30 Gy in five fractions as used in CtE, 42.6% had recurred or died at last follow-up and local relapse directly contributed to the cause of death in 58.3% of those who died [16]. In LRRC and gynaecological cancer, beyond exenteration with all the associated morbidity, there are limited effective alternatives and both quality and quantity of life are entirely dependent on good local control. In an effort to improve local control in patients with LRRC and gynaecological cancer, where the only potentially curative alternative is exenteration, higher ablative doses than those delivered in CtE may have a role. In the guidance, 30 Gy

in five fractions is proposed for oligometastatic lymph nodes from previously irradiated prostate cancer. However, in other tumour types, the guidance offers the option of isotoxic dose escalation, whereby the prescription dose is escalated to a maximum of 45 Gy in five fractions, until the OAR constraints are met. This allows the potential for ablative doses where appropriate, but is designed to maintain an acceptable toxicity risk, as dose prescription is dictated by predefined OAR dose volume constraints, which will be met. This technique has been successfully used in the phase III FLAME study in prostate radiotherapy, improving freedom from disease while maintaining acceptable toxicity [17]. Robinson *et al.* [18] published a planning study looking at the feasibility of isotoxic dose escalation in LRRC and reported the median EQD2 deliverable to 80% or more of the planning target volume increased from 43 Gy to 61 Gy using an isotoxic technique, representing an increase from a median BED of 52 Gy–73 Gy [18]. Thirty-eight per cent of patients achieved an ablative dose, defined as 80 Gy BED ($\alpha/\beta = 10$ Gy) to at least 80% of the planning target volume. Ideal patients for isotoxic treatment do not have any OARs immediately adjacent to the target and clear verification images to ensure OAR movement is identified and accounted for on set.

If centres are keen to contribute to the audit, we would suggest they first seek local approval for the audit from the Divisional Governance Committee and contact the UK SABR Consortium to receive a pack, including the guidance, the PROM questionnaires and a short summary of what data should be collected and documented prospectively. Only centres contributing to the audit will have access to the guidance. Centres would then deliver pelvic SABR re-irradiation using the guidance for the duration of the audit. The audit will run from the publication of this editorial until the data collection planned in late 2024. The data collection will be led by the RCR COQIAC and will be carried out with a specifically developed web-based data collection form. This form will collect data on every patient treated, using these guidelines, over this time period.

We aim to deliver on a truly national project across the UK, to be led and driven by NHS clinicians delivering this service. The ideal remains prospective trials, however the data collected in this audit will allow us to optimise SABR re-irradiation in the pelvis and provide data that will better position us in designing, funding and running prospective trials in future. We welcome comments, questions and contributions, so please get in touch (sabr@profileproductions.co.uk).

Conflicts of Interest

The authors declare no conflicts of interest.

Author Contributions

RM was responsible for conceptualisation and writing. CD, PD, MW and FMCD were responsible for the methodology of the proposed guidelines – audit, review and editing.

MW and FMCD were responsible for supervision in chairs of perspective committees (COQIAC and UK SABR Consortium).

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